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SUB NAME: FUNDEMENTALS OF DATA SCIENCE

Case Study 1 — Employee Salaries

Given the salaries (in ₹ lakhs):

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Mean, Median, and Mode

Mean

[
$$\text{Mean} = \frac{4+5+5+6+7+8+8+9+30}{9}$$
$$= \frac{82}{9}$$
$$= 9.11$$

]

Answer: Mean = 9.11 lakhs

Median

There are 9 values, so the median is the middle (5th) value.

Ordered data:

4, 5, 5, 6, 7, 8, 8, 9, 30

The **5th value = 7**

Answer: Median = 7 lakhs

Mode

The numbers **5** and **8** each occur twice.

Answer: Mode = 5 lakhs and 8 lakhs (Bimodal)

2. Range

[
 $\text{Range} = \text{Maximum} - \text{Minimum}$
 $= 30 - 4 = 26$
]

Answer: Range = 26 lakhs

3. Variance and Standard Deviation (Population)

Mean = **9.11**

Salary (x) (x - $\bar{x}$) (x - $\bar{x}$)²)

4	-5.11	26.12
5	-4.11	16.90
5	-4.11	16.90
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.23

Sum of squared deviations:

[
 $\sum (x - \bar{x})^2 = 512.76$
]

Variance

$$\begin{aligned} &[\\ &\sigma^2 = \frac{512.76}{9} = 56.97 \\ &] \end{aligned}$$

Standard Deviation

$$\begin{aligned} &[\\ &\sigma = \sqrt{56.97} \approx 7.55 \\ &] \end{aligned}$$

Answers:

- **Variance = 56.97**
- **Standard Deviation = 7.55 lakhs**

(If your class uses the sample formula, then variance = 64.10 and standard deviation $\approx$ 8.01.)

4. Identify the Outlier

The salary **30 lakhs** is much larger than the other salaries (which range from 4–9 lakhs).

Answer: 30 lakhs is the outlier.

5. Which Measure Best Represents the Typical Salary?

Median (7 lakhs) is the best measure.

Reason:

- The outlier (30 lakhs) increases the **mean** to **9.11 lakhs**, making it misleading.
 - The **median** is not affected by the outlier and better represents the typical employee salary.
-

2) Case Study 2 — Website Response Time

Given data (in milliseconds):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the Mean and Median

Mean

$$\begin{aligned} &[\\ &\text{\text{Mean}} = \frac{120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 \\ &+ 150 + 300}{10} \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= \frac{1505}{10} = 150.5 \text{ ms} \\ &] \end{aligned}$$

Mean = 150.5 ms

Median

There are 10 values (even number), so the median is the average of the 5th and 6th values.

5th value = 132

6th value = 135

$$\begin{aligned} &[\\ &\text{\text{Median}} = \frac{132 + 135}{2} = 133.5 \text{ ms} \\ &] \end{aligned}$$

Median = 133.5 ms

2. Find the Range

[
 $\text{Range} = \text{Maximum} - \text{Minimum}$
]

[
 $= 300 - 120 = 180 \text{ ms}$
]

Range = 180 ms

3. Calculate the Variance and Standard Deviation

Using the population formula:

Value (x) $(x - \bar{x})$ $((x - \bar{x})^2)$

120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25

Value (x) (x- $\bar{x}$) ((x- $\bar{x}$)²)

300 149.5 22350.25

Sum of squared deviations:

**[
25581.5
]**

Variance

**[
 $\sigma^2 = \frac{25581.5}{10} = 2558.15$
]**

Variance = 2558.15 ms²

Standard Deviation

**[
 $\sigma = \sqrt{2558.15} \approx 50.58$
]**

Standard Deviation ≈ 50.58 ms

4. Determine Whether the Data is Right-Skewed or Left-Skewed

- Mean = 150.5 ms**
- Median = 133.5 ms**

- One very large value (300 ms) pulls the mean upward.

Answer: The data is right-skewed (positively skewed).

5. Why is the Median More Useful Than the Mean?

The response time of 300 ms is an outlier.

- The mean (150.5 ms) is increased by this unusually high value.
- The median (133.5 ms) is not greatly affected by the outlier and better represents the typical website response time.

Answer: The median is more useful because it is less affected by extreme values (outliers) and gives a better measure of the typical response time.

Case Study 3 — Student Performance

Given data (already in ascending order):

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2, and Q3

There are 12 observations (even number).

Q2 (Median)

Median is the average of the 6th and 7th values.

6th value = 52

7th value = 55

[

$$\mathbf{Q_2 = \frac{52+55}{2} = 53.5}$$

]

$$\mathbf{Q2 = 53.5}$$

Q1

First half:

42, 45, 48, 50, 52, 52

Median of first half:

[

$$\mathbf{Q_1 = \frac{48+50}{2} = 49}$$

]

$$\mathbf{Q1 = 49}$$

Q3

Second half:

55, 58, 60, 62, 65, 95

Median of second half:

$$\begin{bmatrix} Q_3 = \frac{60+62}{2} = 61 \end{bmatrix}$$

$$Q_3 = 61$$

2. Calculate the IQR

$$\begin{bmatrix} IQR = Q_3 - Q_1 \end{bmatrix}$$

$$\begin{bmatrix} = 61 - 49 = 12 \end{bmatrix}$$

$$IQR = 12$$

3. Use the $1.5 \times IQR$ Rule to Identify Outliers

Lower Limit

$$\begin{bmatrix} Q_1 - 1.5(IQR) = 49 - 1.5(12) \end{bmatrix}$$

$$\begin{bmatrix} = 49 - 18 = 31 \end{bmatrix}$$

Upper Limit

$$[\\ Q_3 + 1.5(IQR) = 61 + 18 = 79 \\]$$

Values below 31 or above 79 are outliers.

Only 95 is greater than 79.

Outlier = 95

4. Find the Mean and Median

Mean

$$[\\ \text{Sum} = 42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 = 684 \\]$$

$$[\\ \text{Mean} = \frac{684}{12} = 57 \\]$$

Mean = 57

Median

Already calculated:

Median = 53.5

5. Explain How the Outlier Affects the Mean

The value 95 is much higher than the other marks.

- It increases the mean from what would otherwise be closer to the center of the data.**
- The median (53.5) is not greatly affected because it depends only on the middle values.**

Answer: The outlier 95 pulls the mean upward, making it larger than the median. Therefore, the median better represents the typical student performance.

Case Study 4 — E-Commerce Orders

Given data (number of orders over 10 days):

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the Mean, Median, and Mode

Mean

$$\left[\begin{aligned} \text{Mean} &= \frac{18+20+22+22+24+25+26+28+30+35}{10} \end{aligned} \right]$$

$$\left[\begin{aligned} &= \frac{250}{10} = 25 \end{aligned} \right]$$

Mean = 25

Median

There are 10 values, so the median is the average of the 5th and 6th values.

5th value = 24

6th value = 25

$$\left[\begin{aligned} \text{Median} &= \frac{24+25}{2} = 24.5 \end{aligned} \right]$$

Median = 24.5

Mode

The value 22 occurs twice, while all others occur once.

Mode = 22

2. Find the Variance and Standard Deviation

Using the population formula:

x	$x - 25$	$(x - 25)^2$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

$$[\sum (x - \bar{x})^2 = 228]$$

Variance

$$[\sigma^2 = \frac{228}{10} = 22.8]$$

Variance = 22.8

Standard Deviation

$$\begin{aligned} &[\\ &\sigma = \sqrt{22.8} \approx 4.77 \\ &] \end{aligned}$$

Standard Deviation ≈ 4.77

3. Calculate the Coefficient of Variation (CV)

$$\begin{aligned} &[\\ &CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100 \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= \frac{4.77}{25} \times 100 \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= 19.08\% \\ &] \end{aligned}$$

Coefficient of Variation $\approx 19.08\%$

4. What Does the Standard Deviation Tell the Company?

The standard deviation (4.77) indicates how much the daily orders vary from the average.

- A small standard deviation means the daily orders are fairly consistent.

- A larger standard deviation means there is more variation in daily orders.

Answer: Since the standard deviation is about 4.77 orders, the daily orders are fairly close to the average of 25 orders.

5. Is the Order Data Highly or Moderately Variable?

The Coefficient of Variation (19.08%) is below 20%, indicating relatively low variability.

Answer: The order data is moderately variable (fairly consistent), not highly variable.

Case Study 5 — Comparing Two Data Science Models

Given prediction errors:

- **Model A:** 2, 3, 3, 4, 4, 5, 5, 6
- **Model B:** 1, 1, 2, 4, 5, 7, 8, 9

There are 8 observations in each model.

1. Calculate the Mean Error

Model A

$$[$$

$$\text{Mean} = \frac{2+3+3+4+4+5+5+6}{8}$$

$$= \frac{32}{8} = 4$$

$$]$$

$$\text{Mean (A)} = 4$$

Model B

$$[$$

$$\text{Mean} = \frac{1+1+2+4+5+7+8+9}{8}$$

$$= \frac{37}{8} = 4.625$$

$$]$$

$$\text{Mean (B)} = 4.625$$

2. Calculate the Variance (Population)

Model A

$$x \quad x - 4 \quad (x - 4)^2$$

$$2 \quad -2 \quad 4$$

$$3 \quad -1 \quad 1$$

$$3 \quad -1 \quad 1$$

$$4 \quad 0 \quad 0$$

$$4 \quad 0 \quad 0$$

$$5 \quad 1 \quad 1$$

$$x \quad x - 4 \quad (x - 4)^2$$

$$5 \quad 1 \quad 1$$

$$6 \quad 2 \quad 4$$

$$\left[\sum (x - \bar{x})^2 = 12 \right]$$

$$\left[\text{Variance} = \frac{12}{8} = 1.5 \right]$$

$$\text{Variance (A)} = 1.5$$

Model B

$$\text{Mean} = 4.625$$

$$x \quad x - 4.625 \quad (x - 4.625)^2$$

$$1 \quad -3.625 \quad 13.1406$$

$$1 \quad -3.625 \quad 13.1406$$

$$2 \quad -2.625 \quad 6.8906$$

$$4 \quad -0.625 \quad 0.3906$$

$$5 \quad 0.375 \quad 0.1406$$

$$7 \quad 2.375 \quad 5.6406$$

$$x \quad x - 4.625 \quad (x - 4.625)^2$$

$$8 \quad 3.375 \quad 11.3906$$

$$9 \quad 4.375 \quad 19.1406$$

$$[\sum (x - \bar{x})^2 = 69.875$$

]

$$[\text{Variance} = \frac{69.875}{8} = 8.7344$$

]

Variance (B) ≈ 8.73

3. Calculate the Standard Deviation

Model A

$$[SD = \sqrt{1.5} \approx 1.22$$

]

Standard Deviation (A) ≈ 1.22

Model B

$$[SD = \sqrt{8.7344} \approx 2.96$$

]

Standard Deviation (B) $\approx$ 2.96

4. Which Model Has More Consistent Predictions?

A smaller standard deviation means the prediction errors are more consistent.

- Model A: SD $\approx$ 1.22**
- Model B: SD $\approx$ 2.96**

Answer: Model A has more consistent predictions because its prediction errors vary less.

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better?

No.

A lower mean error is important, but it does not always mean the model is better. Other factors should also be considered, such as:

- Variance or standard deviation (consistency)**
- Accuracy**
- Precision and recall**
- F1-score**
- Performance on unseen (test) data**

A model with a slightly lower mean error but very high variability may perform less reliably than a model with slightly higher mean error but much more consistent predictions.
